Events, Entities, and Text

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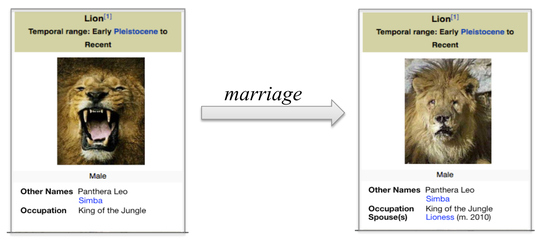
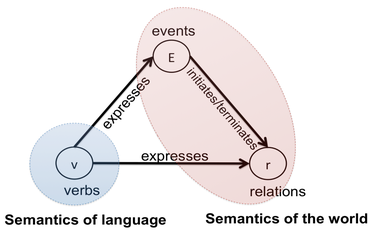
Motivation and Goal

- **Events**, **entities** and **text**: when *state-changing* **events** e.g., *marriage* happen to **entities**, some of the entities may be changed. The change in the entities may be visible from the way the entities are described in **text**.
- We **study** *how* entities *change* as events happen to them based on *how* the entities are *described* in text
- Knowledge base (KB) such as DBPedia (a "database" of Wikipedia), NELL, contains **knowledge** (*relation-value* pairs) about entities in the world
  - Knowledge in KB is useful in areas such as information retrieval, information extraction, and natural language processing
  - Studying how entities change as events happen can be useful for *updating knowledge* in KB
  - For example, when *state-changing* **events** e.g., *marriage* happen to entities, knowledge about their SPOUSE change and need to be updated



- The **goal** of the study is to use what we discover on the correspondences between events, entities and verbs in text to create a resource of verbs containing: [1]
  - Correspondences between verbs and relations in KB (e.g., between the verb "work as" and the relation PROFESSION). This is useful for *populating* knowledge in the KB[1]
  - Correspondences between verbs and *changes* in relations in KB (e.g., between the verb "fire" and the *termination* of the relation PROFESSION). This is useful for updating *knowledge* in the KB[2]

focus in this workshop ➡

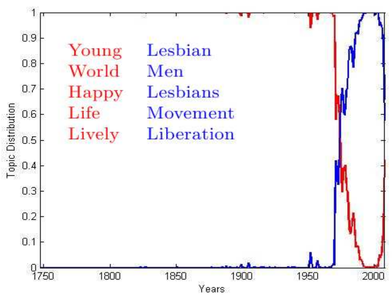


[1] Wijaya, Derry Tanti, Ndapandula Nakashole, and Tom M. Mitchell. "Mining and organizing a resource of state-changing verbs." Proceedings of the Joint Workshop on Automatic Knowledge Base Construction and Web-scale Knowledge Extraction. 2014.

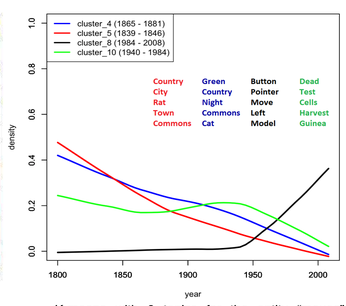
1st Study:

Identify Changes in Entities from Their Mentions in Text

- Tracking the *frequency* of an entity's mentions in a large collection of texts (Google Books NGram [2]) over time can indicate when a *change* is happening to an entity as its frequency changes [3]
- We model and analyze changes that occur to an entity in terms of changes in the *words* that co-occur with the entity in text (Google Books NGram) over time [4]
  - We *cluster* words that co-occur with an entity in 5-grams over time (year-granularity)
  - We use K-means and Topics-over-Time (TOT) clustering [5], an LDA-style topic model that captures how latent structure in the data changes over time
  - We identify there is a change to the entity when two consecutive years are of different clusters
  - We observe that the period of change identified by our model precisely coincides with events in the same period that correspond to the change that occurs
    - For the entity *Iran*, our model identifies a change in the year 1979 with the emergence of a new cluster containing words such as "republic" and "revolution". The year 1979 is also the year of the *Iran* Islamic revolution
    - For the entity *Kennedy*, our model identifies a change in the year 1961 with a cluster containing words such as "senator" transitioning to a new cluster containing words such as "president". The year 1961 is also the year *John F. Kennedy* was elected president



TOT with 2 topics for the entity "gay" shows a cluster transition in 1970



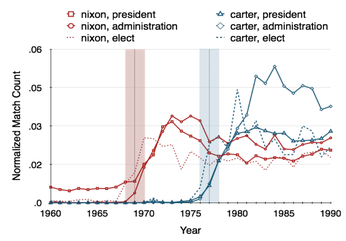
K-means with 2 topics for the entity "mouse" shows cluster transitions in 1940 (mice for animal testing research) and in 1984 (mouse used with personal computers

[2] https://books.google.com/ngrams
[3] Michel, Jean-Baptiste, et al. "Quantitative analysis of culture using millions of digitized books." science 331.6014 (2011): 176-182.
[4] Wijaya, Derry Tanti, and Reyyan Yeniterzi. "Understanding semantic change of words over centuries." Proceedings of the 2011 international workshop on DETecting and Exploiting Cultural diversiTy on the social web. ACM, 2011.
[5] Wang, Xuerui, and Andrew McCallum. "Topics over time: a non-Markov continuous-time model of topical trends." Proceedings of the 12th ACM SIGKDD international conference on Knowledge discovery and data mining. ACM, 2006.

2nd Study:

Analyze the Context of Mentions of Changed Entities in Text

- We construct for each entity e, its *Contextual Temporal Profile* (CTP) at time t, $C_e(t)$ consisting of the context within which e is mentioned – unigrams and bigrams from sentences mentioning e at time t [6]



Frequency of context unigrams surrounding US president entities in Google Books NGram shows a behavior pattern: the rise of "elect", immediately followed by the rise of "administration" and "president"

- We compute change patterns in the *CTP* as the state and change vectors:
  - State vector of an entity e at time t is $C_e(t)$
    - This reflects the *events* happening to the entity e and its *state* at time t
  - Change vector of an entity e at time t is defined as $C_e(t) - C_e(t-1)$
    - This *difference* in context before, at time $t-1$, and after, at time t, reflects the associated *state change* of the entity e at time t

Relation	CTP State Context	Unigrams and Bigrams in CTP Change Vectors
US President	was elected, took office, became president	vice president (-), by president (+), administration (+), senator (-), governor (-), candidate (-)
Best Picture	nominated for, to win, won the, was nominated	best picture (+), hour minute (-), academy award (+), oscar (+), nominated (+), won (+), star (-), best actress (+), best actor (+), best supporting (+)

Example behavior patterns of contextual units (unigrams and bigrams) in the state and change vectors of entities from the ***presidentOf***(*e*, US) and ***winAward***(e, Best Picture) relations in NELL. The (+) and (-) signs indicate rise and fall in mention frequency, respectively. The *state* vectors contain words indicative of events that correspond to the relation such as "was elected", "won". The *change* vectors contain words that change in frequencies corresponding to the event

- We use the state and change vectors to predict *when* an entity undergoes a particular change from the context of its mentions. We use the prediction to *temporally scope* the entity's corresponding relation in KB effectively

[6] Wijaya, Derry, Ndapa Nakashole, and Tom Mitchell. "CTPs: Contextual temporal profiles for time scoping facts via entity state change detection." Proceedings of the Conference on Empirical Methods in Natural Language Processing. Association for Computational Linguistics. 2014.

Approach

We conduct a series of studies that aims to answer the following questions:

- Can we identify changes of entities from the ***context*** within which they are mentioned in text?
- Can we find in the context within which the changed entities are mentioned in text, words that ***trigger*** the changes?
  - Are these words indicative of the event that correspond to the changes?
- Given the trigger verbs (verbs/verbs + prepositions) that we learn to relate to changes corresponding to a particular event, can we ***predict*** when a similar event happens to other entities from the context of their mentions in text?
  - Can we automatically ***update*** our knowledge about the entity based on the corresponding changes?

3rd Study:

Use Context (Verbs) to Predict Changes to Entities

- We use distantly-labeled documents obtained from Wikipedia revision as our training data to learn which verbs (features) result in which changes (labels) [7]

Wikipedia revision of https://en.wikipedia.org/wiki/Kim_Kardashian on 25 May 2014

[7] Wijaya, Derry Tanti, Ndapandula Nakashole, and Tom M. Mitchell. "'A Spousal Relation Begins with a Deletion of engage and Ends with an Addition of divorce": Learning State Changing Verbs from Wikipedia Revision History."

- We use a Maximum Entropy (MAXENT) classifier[3] to estimate the verb weights $\mathbf{w} \in R^{|V|}$ for each change label

Label	Verb
begin-deathdate	+(arg1) die on (arg2), +(arg1) die (arg2), +(arg1) pass on (arg2)
begin-birthplace	+(arg1) be born in (arg2), +(arg1) bear in (arg2), +(arg1) be born at (arg2)
begin-predecessor	+(arg1) succeed (arg2), +(arg1) replace (arg2), +(arg1) join cabinet as (arg2), +(arg1) join as (arg2)
begin-successor	+(arg1) lose **seat** to (arg2), +(arg1) resign on (arg2), +(arg1) resign from post on (arg2)
begin-termstart	+(arg1) be appointed on (arg2), +(arg1) serve from (arg2), +(arg1) be elected on (arg2)
begin-spouse	+(arg1) marry on (arg2), +(arg1) marry (arg2), +(arg1) be married on (arg2), -(arg1) be engaged to (arg2)
end-spouse	+(arg1) file **for divorce** in (arg2), +(arg1) die on (arg2), +(arg1) divorce in (arg2)
begin-youthclubs	+(arg1) start career with (arg2), +(arg1) begin **career** with (arg2), +(arg1) start with (arg2)

Verbs we learn to relate to various changes. The (+) and (-) signs indicate addition and deletion of the verbs from text, respectively.

- Our experiments show that when state-changing verbs are added or deleted from an entity's Wikipedia page text, we can predict the entity's infobox updates with 88% precision and 76% recall
- One compelling application of our state-changing verbs is to incorporate them as triggers in methods for updating existing knowledge bases (KBs), which are currently mostly static, from reading texts e.g., news text

Footnotes

1. We make the verbs learned for various KB (NELL) relations available in: http://www.cs.cmu.edu/~dwijaya/mapping.html
2. We make the verbs learned for various *changes* in KB (Wikipedia) relations available in: http://www.cs.cmu.edu/~dwijaya/postcondition.html
3. We use MALLET implementation of MAXENT: http://mallet.cs.umass.edu

Acknowledgments

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